DBSCAN

Agenda

↳ Core, Border & Noise point

↳ Algo

↳ Hyper. tuning

↳ Time Complexity

DBSCAN — Density Based Spatial Clustering of Application with Noise

K-means

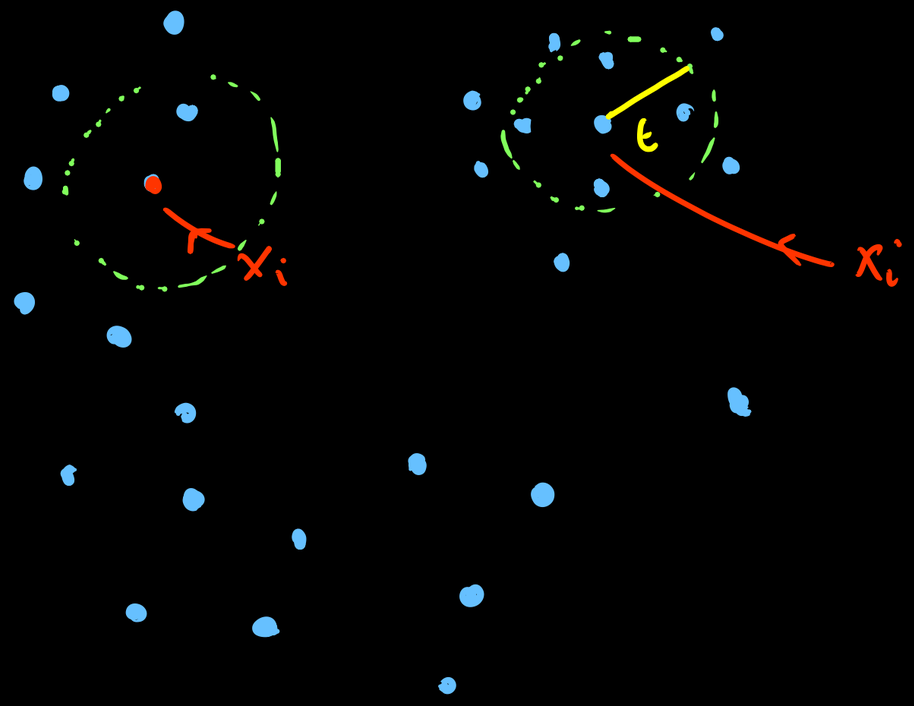
↳ Does not work for non-globular data

↳ Highly sensitive to outliers

Hyperparameter

1) Min_pts = 3

2) ϵ (radius) = 2



$4 \geq 3$

DBSCAN ✓

→ Cone point

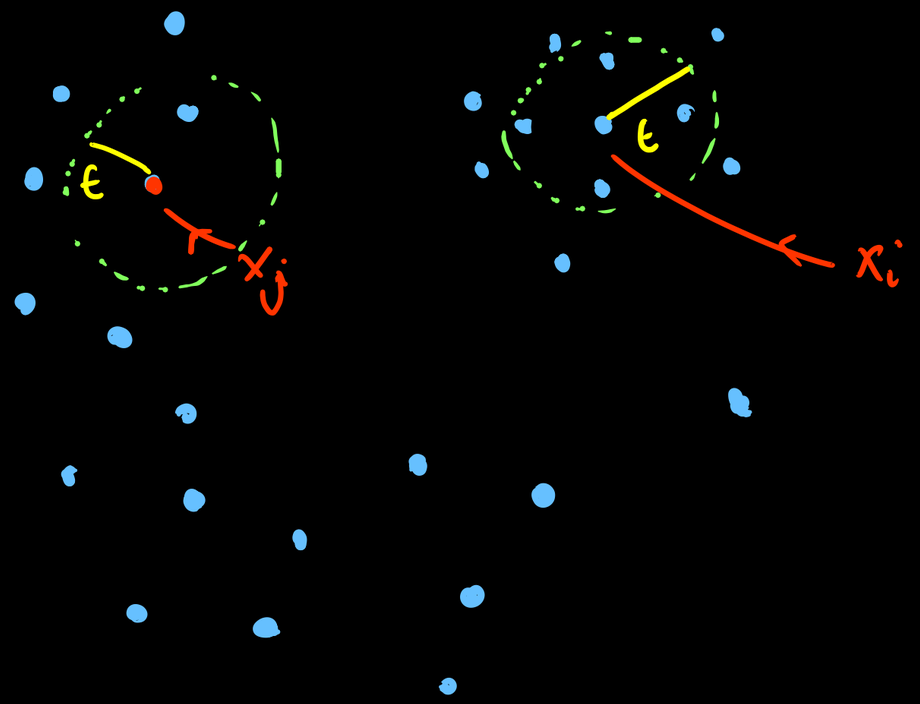
→ Border point

→ Noise point

Hyperparamter

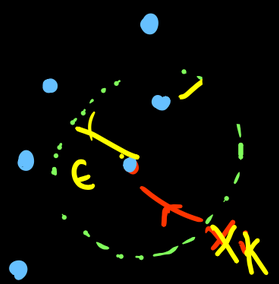
1) Min_pts = 3

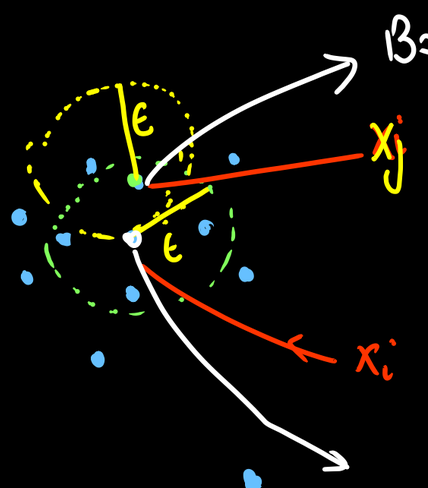
2) ϵ (radius) = 2



Cone point : A point is called cone point if it has atleast min_pts within a radius ϵ !

$4 \geq 3$ = x_i is Cone point





Border point

Core point

Hyperparamter

1) Min-pts = 3

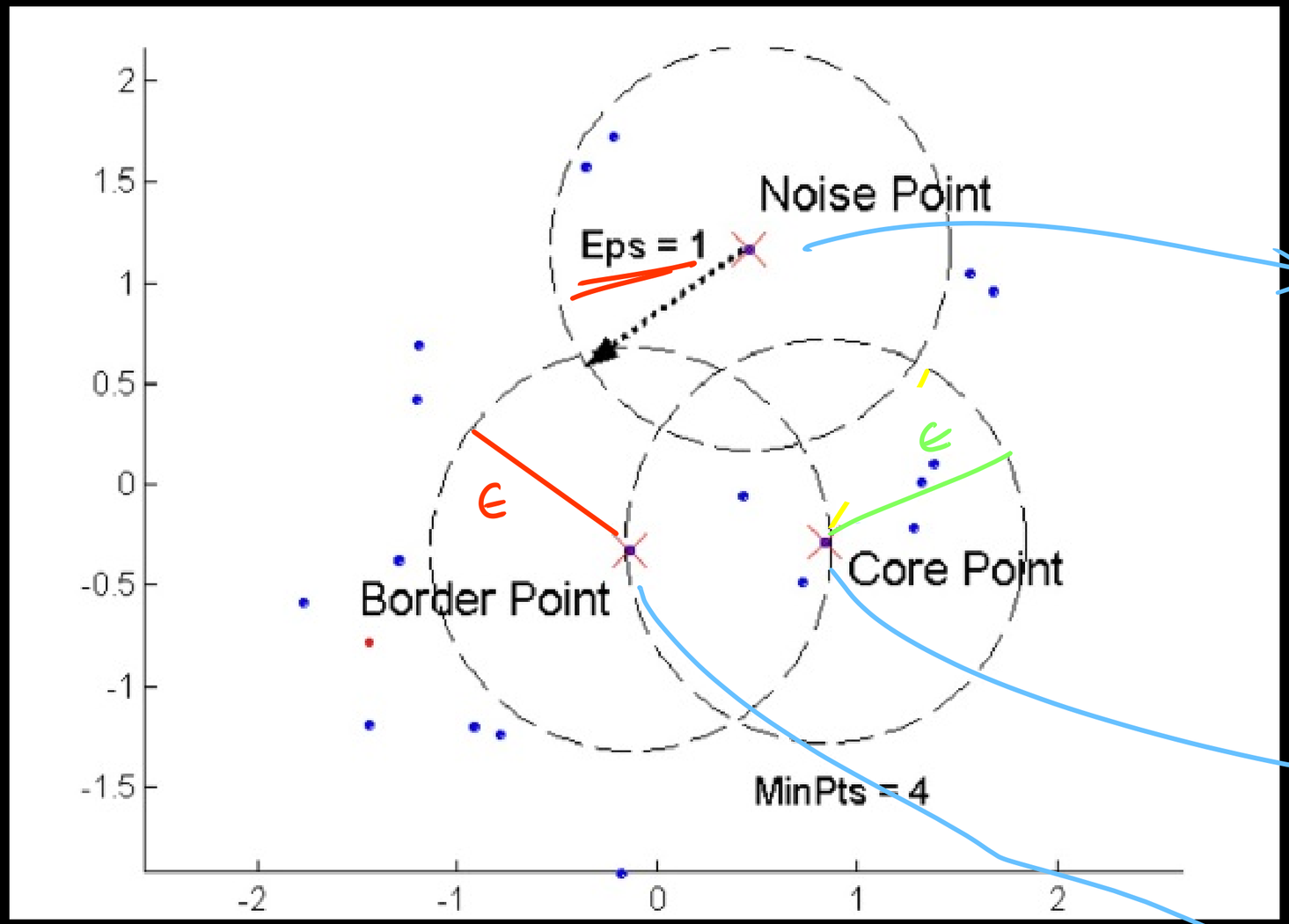
2) ϵ (radius) = 2

Border point → 2 > 3 (not a core point ✓)

↳ Not a core point ✓

↳ Lie in ϵ radius of ✓ atleast one core point

Noise Point → Neither core nor border point



min.pts = 4

$\epsilon = 1$

$2 \geq 4 \rightarrow N_o$

$5 \geq 4 =$ Core Point

X_1

X_2 $3 \geq 4 \rightarrow$ Not a Core point

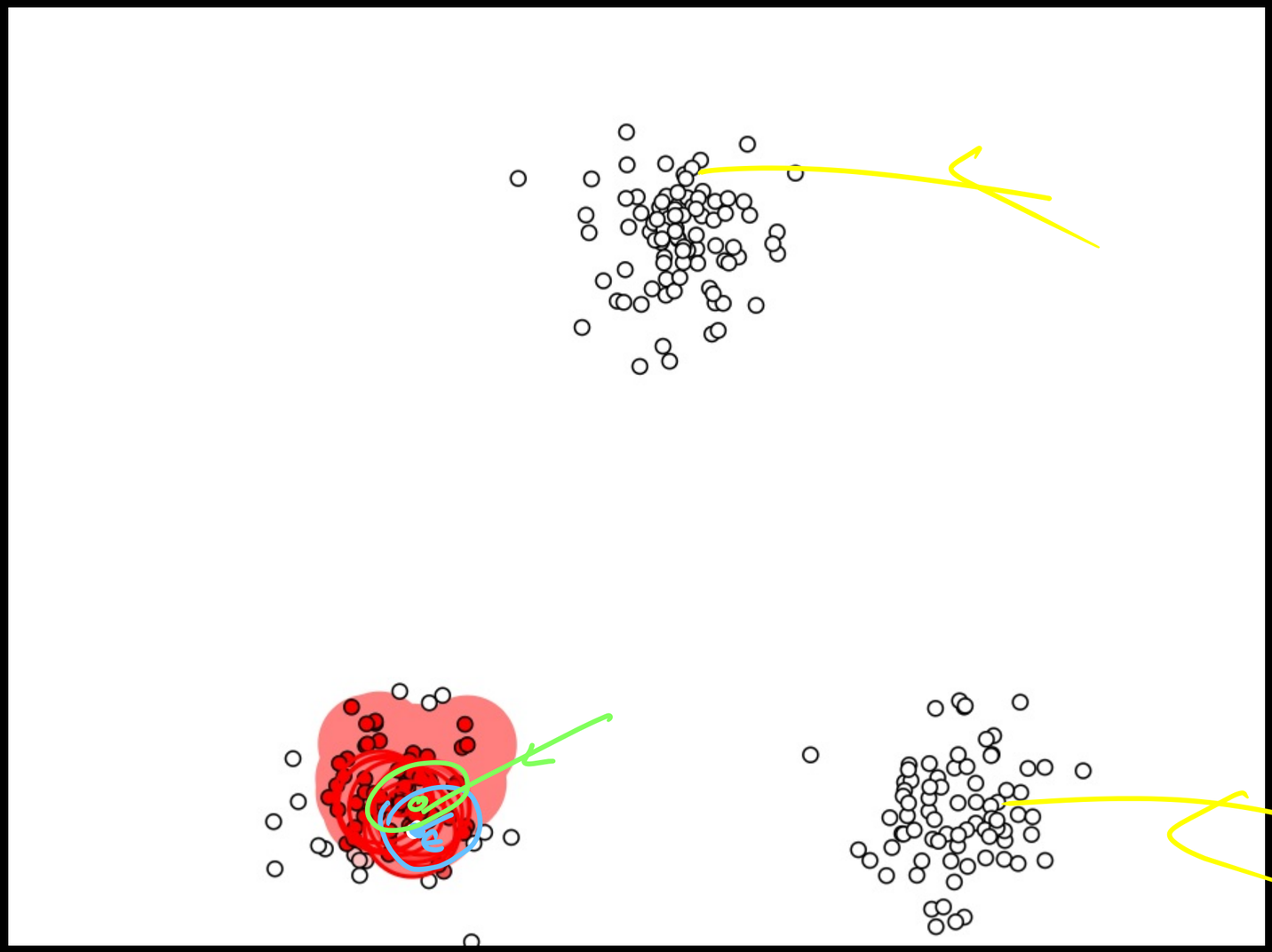
Density at point x_i : No of points within cinde of radius ϵ

Dense Region : A cincle of radius ϵ contain atleast min_pts

Algorithm

Repeat until all point visited

- → Start with random unvisited point
- → Find the No of pts within ε radius
- → If No of pts > min-pts
  - ↳ Core point
- → Expand the cluster (if core point)
  - ↳ Add All the neighbour (ε-radius) to the cluster
  - ↳ If the neighbour is also core point then it neighbour are also added
- →



Hyperparameter

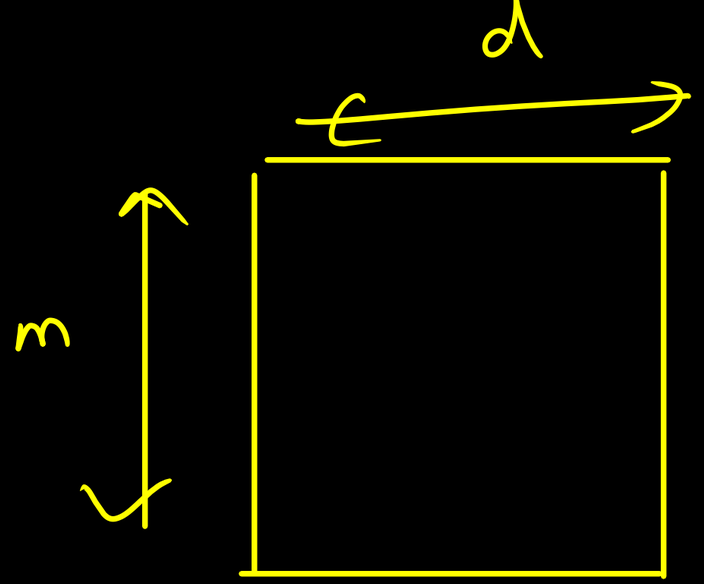
1) Determine min-pts

dimension of data

$\geq d+1$

$min_pts \geq d+1$

$min_pts \approx 2d$



2> Determine ϵ radius

① Calculate K-NN distance for every point

↳ K = min pts

= 4

$x_i \longrightarrow$ K - NN distance

4 - NN distance

② Sort the distance in ascending order

③ Elbow point

① Calculate K-NN distance for every point

↳ K = min pts

= 4

$x_i \longrightarrow$ K - NN distance

4 - NN distance

② Sort the distance in ascending order

③ Elbow point

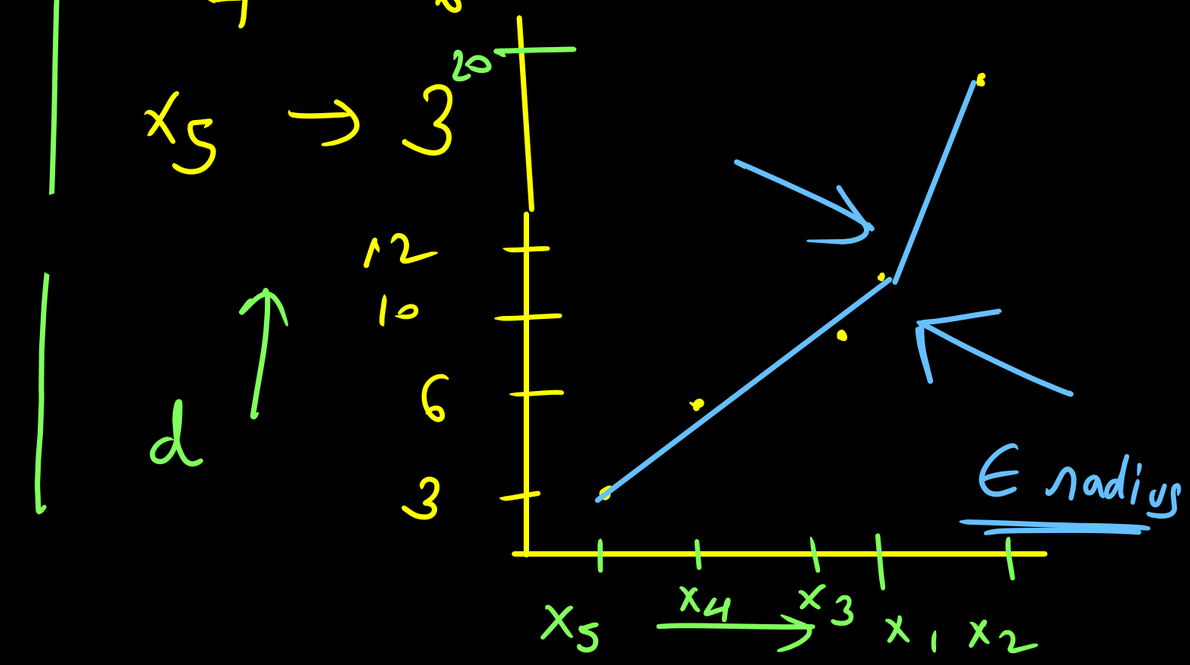
$x_1 \longrightarrow 10$

$x_2 \longrightarrow 20$

$x_3 \longrightarrow 12$

$x_4 \longrightarrow 6$

$x_5 \Rightarrow 3$



→ Break until 22:25 PM

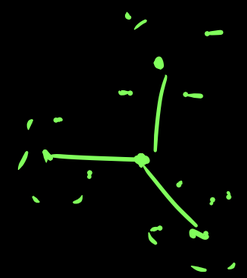
When are two points considered "density connected" in DBSCAN?

4 options

Active Duration(Most preferred: 30 seconds)

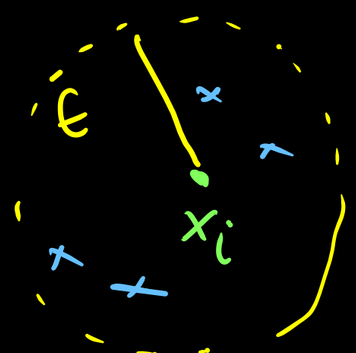
Appears for	30 Secs

A	When they are connected by an edge
B	When they are both noise points in the dataset
C	When they are core points and there exists a sequence of density edges connecting them
D	When they are part of the same cluster



Time Complexity

For each point, identify the points that lie within ϵ radius



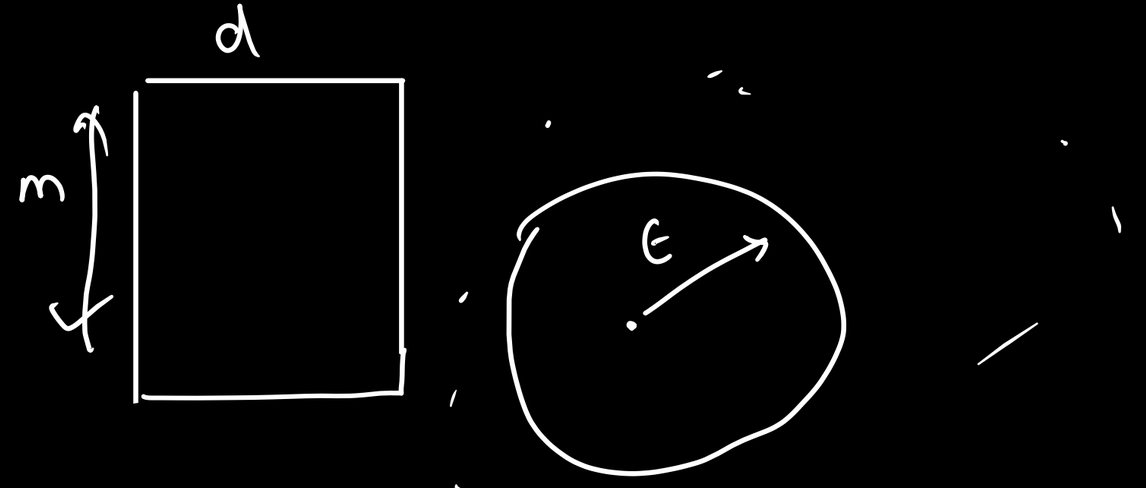
↳ $O(n)$

n points ⟶ $O(n * n) = O(n^2)$ ✓

Range query ⟶ $O(n \log n)$

→ ϵ radius

→ min_pts = 4



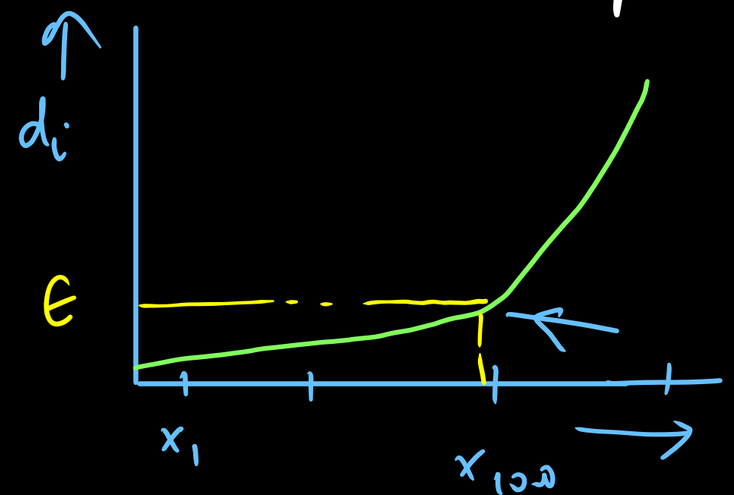
x_1 ⟶ 4th NN d_1

x_2 ⟶ d_2

⋮ ⋮

x_m ⟶ d_m

No of neighbours ≥ min_pts
ϵ radius ⟶ Core point



K-NN

4 = min pts

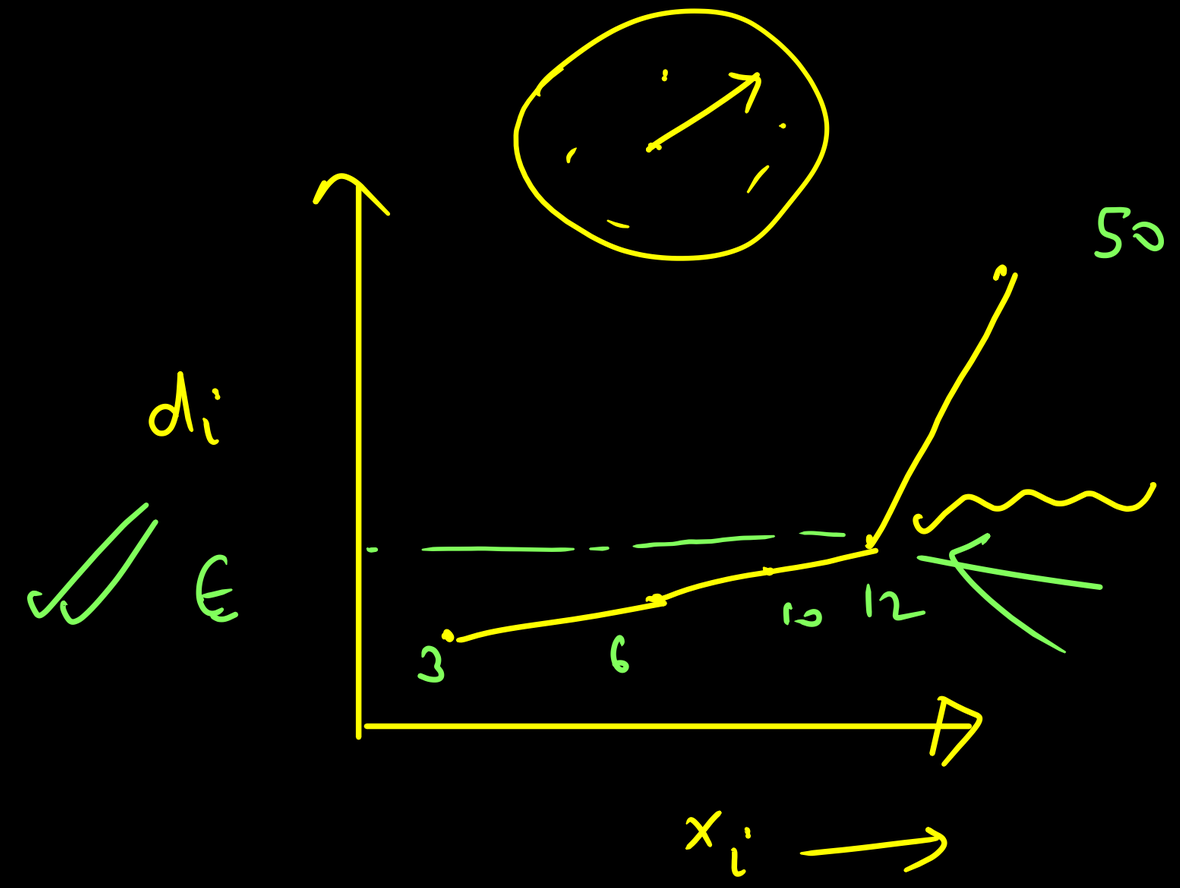
$x_1 \rightarrow 10$

$x_2 \rightarrow 6$

$x_3 \rightarrow 12$

$x_4 \rightarrow 50$

$x_5 \rightarrow 3$



$\rightarrow$ 3, 10, 6, 12, 50

→ Break until 22:30 PM

Time Complexity

1. For each point, identify the points that lie within ϵ radius $\rightarrow O(n)$

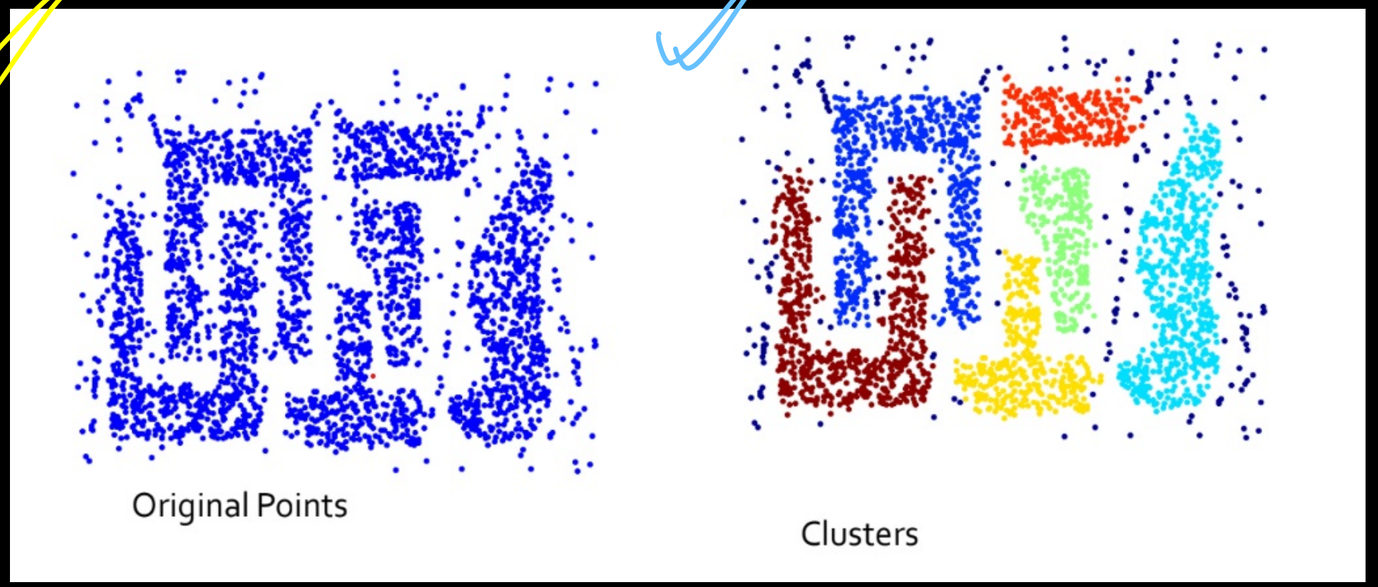
n points $\rightarrow O(n * n) = O(n^2)$

Range query

n points $\rightarrow O(n \log n)$

Advantage →

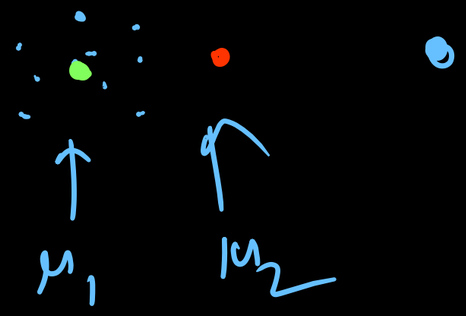
Works well for Non-globular structure



→ Resistant to Noise (outliers)

↳ Seperate Noise point for outlier

↳ Outlier does not take part in clusters



← Outlier affected

K-means

Limitations

→ Slow

→ Sensitive to hyperparameter

$\epsilon = 0.5$, min = 4

$\epsilon = 0.4$, min-pts = 4

Figure 8. DBScan results for DS1 with MinPts at 4 and Eps at (a) 0.5 and (b) 0.4.

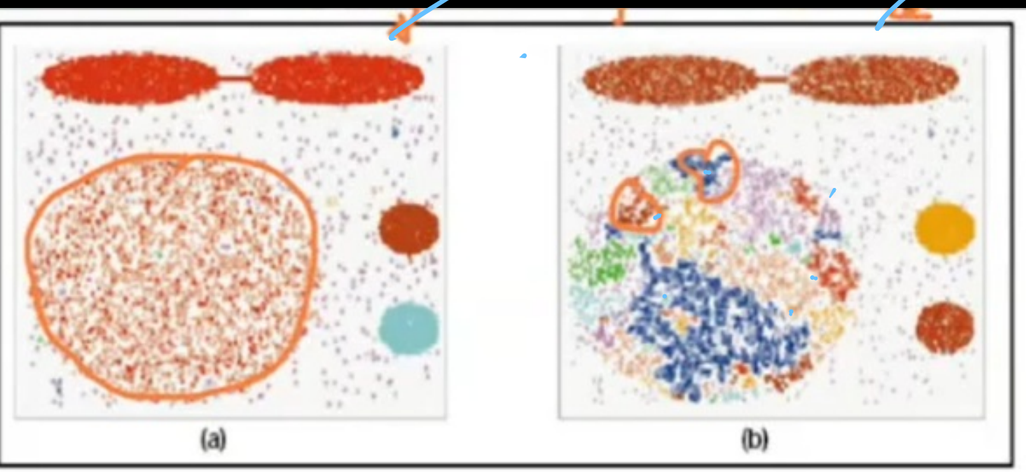
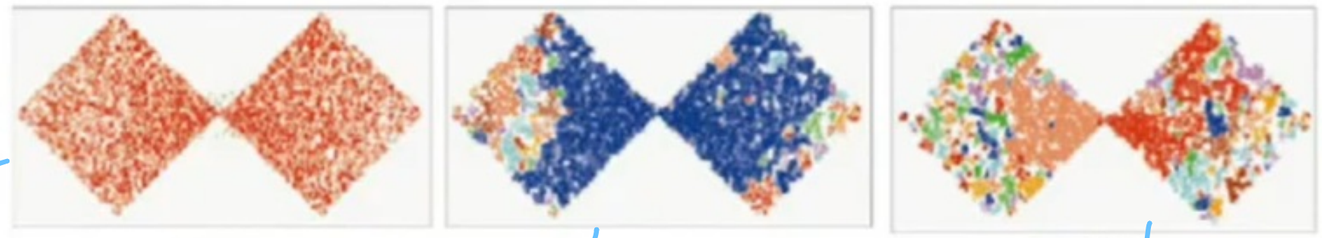


Figure 9. DBScan results for DS2 with MinPts at 4 and Eps at (a) 5.0, (b) 3.5, and (c) 3.0.



e = 5

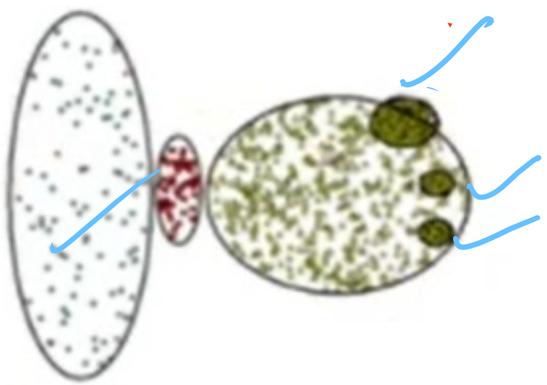
3.5

3

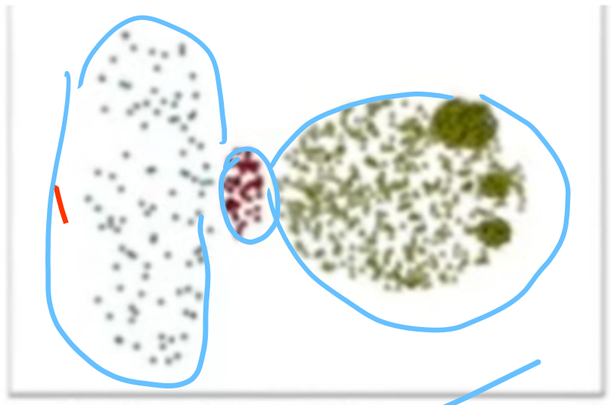
→ Data of diff. density

ϵ. radius

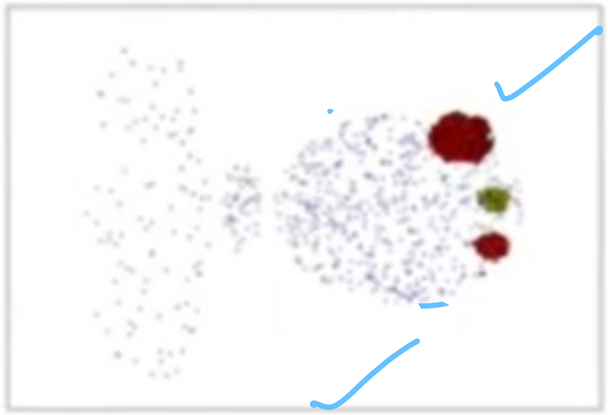
min - pnts



Original Points



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

- Varying densities
- High-dimensional data

- → Linear / Logistic
- → Tree (DT, RF, XGB, LightGBM)
- ⟷ KNN ($O(N^2)$) → LSH

- → PCA
- → Clustering (K-means, DBSCAN, Hierchial)